CHAEWON YOON

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RESEARCH FOCUS

Efficient and reliable machine learning under computational and numerical constraints; inverse vision, fractional-PDE neural operators, low-precision inference, mechanistic interpretability, and causal model evaluation.

EDUCATION

University of Wisconsin-Madison | B.S. in Data Science (in progress) *Expected May 2028*

Transferred Fall 2026. Coursework includes linear algebra and optimization, data-science modeling, discrete mathematics, calculus, probability, and applied statistics.

University of Wisconsin Oshkosh | Undergraduate Study *Sep. 2024-May 2026*

Faculty-mentored independent studies in neural methods for PDEs, numerical mathematics, stochastic processes, and vibration modeling.

ACCEPTED AND PEER-REVIEWED PUBLICATIONS

- Yoon, C. *The Measurement Operator Changes Which Bits Matter: Operator-Conditioned Mixed Precision for Inverse Vision.* IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027), Algorithms Track. **Accepted; proceedings forthcoming. Manuscript**
- Yoon, C. *EvidenceLens: A Declarative Evidence-State Interface for Human-AI Visual Inspection.* ACM UIST Adjunct 2026, poster publication. **DOI** | **Paper**

COAUTHORED MANUSCRIPT UNDER REVIEW

- Jeong, J., Laroza, A. J., **Yoon, C.**, Pillai, A., Kokati, P., Hoang, C., and Hong, S. *One Head Away from Language Confusion: Single Attention Heads Across Model Families.* ACL Rolling Review, October 2026 cycle, long paper. **Under review.**

SELECTED INDEPENDENT RESEARCH MANUSCRIPTS

- Yoon, C. *Conditional Precision Bottlenecks and Quantized Defect Correction in Fractional Neural Operators.* 2026. **Preprint** | **Code** | **Software DOI**
- Yoon, C. *Transition-Preserving State Tracking via Definition-Grounded Compilation.* 2026. **Preprint** | **Code**
- Yoon, C. *Evidence Is Not Free: Constrained Template Evidence Acquisition for Risk-Aware Governance.* 2026. **Preprint** | **Code**

MAJOR INDEPENDENT AND FACULTY-MENTORED RESEARCH

Operator-Conditioned Mixed Precision for Inverse Vision | Independent *2026*

- Introduced Operator-Susceptibility Matching (OSM), estimating stage-specific numerical-precision sensitivity from calibrated sensing geometry rather than exhaustively searching precision schedules per sensing operator. **Research repository**
- In an eight-stage ProxSCI-8 experiment spanning 256 binary schedules, OSM attained **2.7% mean normalized operator-visible regret** versus **6.3%** for the best global-static schedule, at a three-stage high-precision budget. A distinct native A100 timing evaluation measured **16.4 ms** per reconstruction versus **23.6 ms** for the FP16 reference (**1.44x speedup**). The quality and timing studies used different precision protocols.

Scientific ML and Fractional-PDE Neural Operators *Sep. 2025-Jul. 2026*

Faculty mentor: Kamran Kazmi, Professor of Mathematics, University of Wisconsin Oshkosh

- Developed Quantized Defect Correction for fractional-Darcy Fourier neural operators; tested Q6 weight and activation interventions on separate OOD holdouts after validation-selected checkpoints and tracked physics residuals alongside prediction error. **Code and protocols**
- Designed matrix-free fractional-operator and selected-rank LOBPCG experiments at multiple resolutions, with frozen validation choices and a separate A100 repeated-query hardware audit. Each evaluation uses its documented checkpoint and workload.

Stochastic Processes and Vibration Modeling *Sep. 2025-May 2026*

Faculty mentor: Ahmed O. Nasif, Associate Professor of Electrical Engineering Technology, University of Wisconsin Oshkosh

- Implemented simulations for ODEs and multi-degree-of-freedom vibration systems, including Hamiltonian formulations, matrix/eigenvalue analysis, physics-constrained neural models, and numerical error/convergence testing. **Presentation DOI**

MULTILINGUAL LLM AND MECHANISTIC INTERPRETABILITY RESEARCH

Single-Head Language Confusion | Coauthored ARR Research *2026*

- Coauthored a study of 12 instruction-tuned models from seven series (1B to 7B parameters), testing attention-head zero and mean ablations against FLORES-200 continuations and Language Confusion Benchmark replies. In ten models a single head had a disproportionately large language-adherence effect.
- Seven selected heads shifted 8.0%-51.5% of 2,500 FLORES continuations out of the expected language; other heads in the same layer produced at most 1.4%. Head-value substitution redirected 84%-99% of scored non-English monolingual LCB replies into a replacement language in five models.

English-to-Spanish LLM Circuits | Merged Research Contributions *2026*

- Implemented Qwen2.5 and Llama-3.2 head discovery, exact activation patching, and source-to-destination path validation. Both studies used 50-example held-out FLORES checks and 100 layer-matched random-head controls for selected head sets. **Qwen PR** | **Llama PR**
- Added reproducible discovery scripts and evaluated frozen Qwen2.5 head-set transfer to the corresponding instruction-tuned checkpoint under a fixed raw-prefix protocol. **Transfer PR**

First-Token Broadcasters / LIHA | Merged Research Contributions *2026*

- Compared mean and zero ablation of 144 GPT-2 attention heads across 2,500 language prompts; tested downstream attention redistribution and extended exploratory mechanism analyses to Qwen2.5 and Gemma-3. **Ablation PR** | **Cross-model PR**

TransformerLens-FDR | Merged Mechanistic Analysis *2026*

- Compared ABC and name-swap corruption on GPT-2 Small indirect-object identification, examining 144 heads. Observed Spearman correlation 0.6224 across absolute head effects and 6 shared heads in the top-nine rankings. **Merged PR**

Station House | LLM Self-Attribution Evaluation *2026*

- Merged an OLMo 3 7B Instruct SFT identity-spoofing pilot: nine prompt conditions, three starter prompts each, 27 recorded conversations and three judge variants (original, blinded, identity-aware). **Merged pilot**
- Proposed a separate 10-response, response-only annotation packet to support blinded review, explicitly distinguishing that protocol from conversational grading; the annotation PR is still open. **Open PR**

RESEARCH SOFTWARE AND OPEN-SOURCE ENGINEERING

AladdinAI | Merged Dataset, Testing and Reliability Work *2026*

- Implemented deterministic, session-grouped train/validation/held-out splits (70/15/15 default) and immutable export versions; this prevents multiple turns in the same recorded session from crossing partitions. **Dataset PR**
- Fixed JSON-aware tool-error detection with five offline recovery tests; corrected OOXML exact-element matching and inline-image ordering in DOCX conversion. **Recovery PR** | **Table parsing** | **Inline images**
- Replaced 31 duplicated page-error implementations with a shared frontend component; the inline-image repair was recognized as a XERJ Corpus Hub reference contribution. **Frontend PR** | **XERJ**

CausationEntropy | Merged Statistical Software Contributions *2026*

- Added regression checks for true lag-one causal edge recovery and corrected the default kNN conditional mutual information distance metric. Proposed a one-correlation-matrix Gaussian CMI optimization, still open for review. **Recovery test** | **Metric fix** | **Open optimization**

SandBoxed | LLM Behavior and Benchmark Tooling *2026*

- Developed a six-prompt provider benchmark exporting generation timings and validation outcomes; added 51 regression tests for prompt-specific behavior restrictions and unsupported actions. **Benchmark** | **Tests**

CURRENT RESEARCH AFFILIATIONS AND FELLOWSHIP

Wisconsin AI Safety Initiative (WAISI) | Technical AI Safety Fellow *Oct. 2026-present*

• Participating in Technical Track seminars and weekly readings on AI safety research and model evaluation. **Program**

Neural Representation Dynamics Lab / UW-Madison Research Group *2026-present*

Collaborating in Professor Grigoris Chrysos's research group on diffusion-based reconstruction.

EMPLOYMENT, TUTORING AND MENTORING

College Readiness Mentor (Mathematics) | UW-Madison PEOPLE Program *Oct. 2026-present*

• Provide after-school math help and college-readiness mentoring for students at La Follette High School. **Program**

Homepage Programmer | University of Wisconsin Oshkosh *May-Aug. 2026*

Summer programming appointment working on the university homepage.

IT Support Desk | University of Wisconsin Oshkosh *Summer 2026*

Provided user-facing student IT support during the summer term.

Mathematics Tutor | Center for Academic Resources, UW Oshkosh *Sep. 2025-May 2026*

• Tutored roughly 10-15 undergraduate students weekly in calculus, linear algebra, precalculus, and college algebra.

ADDITIONAL COLLABORATIVE EVALUATION PROJECTS

AgentGuard | AI Agent Security Evaluation *2026*

• Merged a five-scenario evaluation matrix that separates agent-side approval checks from observe/enforce guard modes and uses an independent oracle and trace replay to assess unauthorized refunds. **Merged PR**

Mammography Robustness Study | Model Evaluation Infrastructure *2026*

• Merged image-level and breast-level prediction exporters that combine CC/MLO views and validate label consistency, record coverage, and patient-level splits; tests used synthetic records, not patient images. **Merged PR**

PRESENTATIONS

• *Placement-Aware Quantization for Fractional-PDE Neural Surrogates.* WiSys SPARK Symposium, July 2026. Poster. **DOI**

• *Numerical Stability and Convergence of Physics-Constrained Neural Networks in Vibration Modeling.* Pi Mu Epsilon Undergraduate Regional Mathematics Conference, 2025. Oral presentation. **DOI**

HONORS, LEADERSHIP AND TECHNICAL SKILLS

• **Culver Business Model Competition, first place**, CurbMap 360 (2025); **Calculus Prize Exam winner** (2025); Putnam Mathematical Competition participant (2024). **Project**

• AI Club: Vice President (Spring 2026) and Treasurer (Spring 2025-May 2026); led paper discussions and technical workshops.

Programming and tools: Python, PyTorch, NumPy, SciPy, pandas, scikit-learn, TensorFlow, Git/GitHub, LaTeX.

Methods: numerical PDEs, neural operators, quantization, mixed precision, causal activation patching, controlled ablations, bootstrap intervals, matched random controls, regression tests.